

Title:
Fraud Detection using Machine Learning

Name: Nikhil Anand

Branch: Electronics & Communication
Engineering

Year: Final Year (2026)

Machine Learning Intern

Internship ID: UMID10122574110

Acknowledgement

I would like to express my sincere gratitude to **Unified Mentor** for providing me with the opportunity to work on this internship project titled “*Fraud Detection using Machine Learning*.” This project has helped me gain practical knowledge and hands-on experience in the field of machine learning.

I would also like to thank my mentors at Unified Mentor for their continuous guidance, support, and valuable suggestions throughout the duration of this project. Their insights played a crucial role in the successful completion of this work.

Abstract

This project focuses on detecting fraudulent transactions using machine learning techniques. With the increasing number of online transactions, fraud detection has become essential. The model analyses transaction data and classifies whether a transaction is fraudulent or legitimate. Various machine learning algorithms are used to improve prediction accuracy and reduce false positives.

Table of Contents

➤ Acknowledgement	1
➤ Abstract	2
➤ Introduction	3
➤ Problem Statement	4
➤ Objectives	5
➤ Dataset Description	6
➤ Methodology	7
➤ Data Preprocessing	8
➤ Exploratory Data Analysis (EDA)	9
➤ Model Building	10
➤ Algorithms Used	11
➤ Model Evaluation	12
➤ Results and Discussion	13
➤ Visualization	14
➤ Challenges Faced	15
➤ Future Scope	16
➤ Conclusion	17
➤ References	18

Introduction

Fraud detection is an important challenge in today's digital world due to the rapid growth of online transactions. Fraudulent activities can cause significant financial losses, and traditional rule-based systems often fail to detect complex and evolving fraud patterns. Machine learning provides an effective solution by automatically identifying patterns in data and detecting suspicious transactions.

This project focuses on building a machine learning model to classify transactions as fraudulent or legitimate using historical data. The dataset includes features such as transaction amount, time, and other relevant attributes. Since fraud datasets are highly imbalanced, special attention is given to improving the detection of fraudulent cases.

The project involves data preprocessing, feature scaling, and model training using algorithms like **Logistic Regression, Decision Tree, and Random Forest**. The models are evaluated using **precision, recall, and F1-score**, which are more suitable for imbalanced data.

The final model helps in accurately identifying fraudulent transactions and can assist in reducing financial risks.

Problem Statement

With the rapid growth of digital payments, online banking, and e-commerce platforms, the number of fraudulent transactions has increased significantly. Detecting such fraudulent activities is a major challenge for financial institutions, as fraudsters continuously evolve their techniques to bypass traditional security systems. Conventional rule-based methods are often ineffective because they rely on predefined patterns and fail to adapt to new and unseen fraud behaviours.

Another key challenge in fraud detection is that the dataset is highly imbalanced, where fraudulent transactions represent only a very small fraction of the total data. This imbalance makes it difficult for standard machine learning models to accurately identify fraud cases, as they tend to favour the majority class (non-fraudulent transactions). As a result, achieving high accuracy alone is not sufficient, and there is a need to focus on metrics like precision, recall, and F1-score.

The objective of this project is to develop an efficient and reliable machine learning model that can accurately classify transactions as fraudulent or legitimate based on available features. The system should be capable of minimizing false negatives (undetected fraud) while also reducing false positives to avoid unnecessary alerts. By leveraging machine learning techniques, the project aims to improve fraud detection performance and contribute to enhanced financial security.

Dataset Description

The dataset used in this project is a **simulated transaction dataset** designed to represent both legitimate and fraudulent financial transactions. It contains essential transaction-level information and is structured to support the development and evaluation of fraud detection models.

Fraud cases in the dataset are generated based on predefined scenarios. Transactions with amounts above a certain threshold are marked as fraudulent to create clear patterns. In another scenario, specific terminals are randomly selected, and all transactions from those terminals over a period are labelled as fraud. Additionally, certain customers are targeted, where a portion of their transactions are altered to higher values to simulate real-world fraud such as misuse of credentials.

The dataset includes key features such as transaction ID, date and time, customer ID, terminal ID, and transaction amount. The target variable indicates whether a transaction is fraudulent or legitimate. These features help in analysing transaction behaviour and building effective machine learning models.

As the dataset is simulated, it provides a controlled environment to test and evaluate fraud detection techniques. It allows better understanding of model performance and helps in identifying different fraud patterns effectively.

Methodology

The project follows a structured approach:

- Data Collection
- Data Preprocessing
- Exploratory Data Analysis
- Feature Engineering
- Model Training
- Model Evaluation

Data Preprocessing

Data preprocessing is an essential step in building an effective fraud detection system, as raw data is often incomplete, inconsistent, and not suitable for direct analysis. In this project, several preprocessing techniques were applied to prepare the dataset for machine learning models.

Initially, the dataset was checked for missing values and duplicates. Missing values were handled appropriately, and duplicate records were removed to ensure data quality and consistency. Since fraud detection datasets are highly imbalanced, where fraudulent transactions are significantly fewer than legitimate ones, techniques such as oversampling, under sampling, or SMOTE were considered to improve model performance.

Feature scaling was applied to normalize numerical features like transaction amount, ensuring that all features contribute equally to the model. Categorical variables, if present, were converted into numerical form using encoding techniques such as label encoding or one-hot encoding.

The date and time feature was further processed to extract useful information such as transaction hour or day, which helps in identifying unusual patterns. Additionally, feature selection was performed to retain only the most relevant attributes and reduce model complexity.

Finally, the dataset was split into training and testing sets, typically using an 80:20 ratio, to evaluate the model's performance on unseen data.

Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) is performed to understand the structure and patterns present in the dataset before applying machine learning models. It helps in identifying important features, detecting anomalies, and gaining insights into transaction behaviour. In this project, the distribution of transaction amounts was analysed to observe how values vary between normal and fraudulent transactions.

A comparison between fraudulent and non-fraudulent transactions was carried out to understand class imbalance and identify distinguishing patterns. Fraud cases were found to be significantly fewer, highlighting the need for careful model evaluation.

Correlation analysis was also performed to study the relationships between different features and identify the most influential variables. Various visualization techniques such as histograms, bar charts, and heatmaps were used to represent the data clearly.

EDA plays a crucial role in improving model performance by providing meaningful insights and guiding feature selection.

Model Building

Model building is a crucial step in developing an effective fraud detection system. In this project, multiple machine learning algorithms were implemented to classify transactions as fraudulent or legitimate. The pre-processed dataset was used to train different models, including **Logistic Regression, Decision Tree, and Random Forest**.

Initially, Logistic Regression was applied as a baseline model due to its simplicity and interpretability. Then, a Decision Tree model was used to capture non-linear relationships and decision rules within the data. Finally, the Random Forest algorithm, an ensemble learning method, was implemented to improve prediction accuracy and reduce overfitting by combining multiple decision trees.

Each model was trained using the training dataset and tested on unseen data to evaluate its performance. Hyperparameters were adjusted to optimize the models and achieve better results.

The goal of model building was to identify the most suitable algorithm that can effectively detect fraudulent transactions while minimizing errors, especially in an imbalanced dataset scenario.

Algorithms Used

➤ **Decision Tree**

- Captures non-linear relationships
- Provides clear decision rules

➤ **Random Forest**

- Ensemble method using multiple trees
- Reduces overfitting
- Provides higher accuracy

- Models were compared to select the best-performing algorithm for fraud detection

Model Evaluation

Models are evaluated using:

- Accuracy
- Precision
- Recall
- F1 Score

Since the dataset is imbalanced, precision and recall are more important than accuracy.

Results and Discussion

The models were compared based on performance metrics. Random Forest performed better than other models due to its ability to handle complex patterns and reduce overfitting.

- Accuracy: 99.17%
- Precision: 58%
- Recall: 32%
- F1 Score: 41%

The final XGBoost model achieved high accuracy and significantly improved precision, indicating reliable identification of fraudulent transactions with fewer false positives. Although recall is moderate, the model maintains a good balance between detecting fraud and minimizing false alarms, as reflected in the improved F1-score from before.

VISUALIZATION

Data visualization is an essential part of this fraud detection project as it helps in understanding the structure and patterns present in the dataset. Various visualization techniques were used to analyze transaction data before building machine learning models. Initially, bar graphs and count plots were created to observe the distribution of fraudulent and non-fraudulent transactions. These plots clearly showed that the dataset was highly imbalanced, with very few fraud cases compared to normal transactions.

Histograms were used to study the distribution of transaction amounts and other numerical features. This helped in identifying skewness and variations in the data. A correlation heatmap was also generated to analyze relationships between different features, which assisted in selecting the most relevant variables for model training.

Box plots were used to detect outliers and understand how fraudulent transactions differ from legitimate ones. After model training, a confusion matrix was plotted to evaluate the classification performance visually. Additionally, an ROC curve was used to measure the model's ability to distinguish between fraud and non-fraud cases.

Overall, visualization helped in better decision-making, improved feature selection, and enhanced model evaluation.

CHALLENGES FACED

During the development of the fraud detection system, several challenges were encountered that required careful handling and problem-solving. One of the primary challenges was dealing with the highly imbalanced dataset. Since fraudulent transactions were very rare compared to normal transactions, the model initially showed bias towards predicting non-fraud cases. To overcome this, techniques such as oversampling, undersampling, and class weight adjustment were applied.

Another challenge was data preprocessing. The dataset required cleaning, normalization, and transformation to make it suitable for machine learning models. Ensuring that all features were properly scaled and relevant was a time-consuming process. Feature selection also posed difficulties, as choosing the wrong features could negatively impact model performance.

Model selection and tuning were also challenging tasks. Different algorithms produced different results, and it was necessary to compare them carefully. Hyperparameter tuning required multiple iterations and increased computational time.

Additionally, balancing precision and recall was difficult, as reducing false positives without missing actual fraud cases required fine adjustments. Despite these challenges, consistent testing and improvements led to a reliable fraud detection model.

CONCLUSION

This project successfully demonstrates the use of machine learning techniques in detecting fraudulent transactions. By analyzing transaction data and applying various algorithms, an efficient fraud detection model is developed.

The project highlights the importance of data preprocessing, feature selection, and model evaluation in building accurate models. Handling class imbalance is a key factor in improving model performance.

Among the implemented models, ensemble methods such as Random Forest show better results due to their ability to handle complex data patterns.

The developed model can help financial institutions detect fraud more effectively and reduce financial losses. It also improves customer trust by ensuring secure transactions.

FUTURE SCOPE

The fraud detection system can be further improved by using advanced techniques such as deep learning and neural networks. These methods can capture complex patterns in data and improve detection accuracy.

Real-time fraud detection systems can also be developed to identify fraud instantly during transactions. Integration with big data technologies can help handle large-scale datasets.

Feature engineering and the use of additional data sources can further enhance model performance. Continuous model training with new data can help adapt to evolving fraud patterns.

REFERENCES

1. Scikit-learn Documentation – Machine Learning Library for Python
2. Unified Mentor – Fraud Detection Dataset (Provided for Project Work)
3. Research Papers on Fraud Detection using Machine Learning
4. Python Official Documentation
5. Data Science Journals and Articles